

© Brendan Choi 2021

B. Choi, *Introduction to Python Network Automation*

https://doi-org.ezproxy.sfpl.org/10.1007/978-1-4842-6806-3_1

1. Introduction to Python Network Automation

Brendan Choi¹

(1) Sydney, NSW, Australia

This chapter serves as a primer to this book and discusses what it feels like to be an IT professional in today's IT industry. We will also discuss the different enterprise IT engineering domain groups and their responsibilities. The chapter then compares each domain group's weaknesses and strengths and draws up a working study plan that serves as the book's foundation. This chapter will also discuss why you might want to learn a programming language, in our case, Python, since this is your primary reason for purchasing this book and get started on your network automation journey. Finally, we will discuss the minimum PC/laptop requirements to build a fully working Python/Linux/network automation lab on a single PC. All lab machines, Linux servers, routers, and switches will be installed on a single PC/laptop using a recommended software set.

**Suggested difficulty level only.*

**This level may vary based on the reader's experience.*



Laying the Foundation

In recent years, network programmability concepts have been taking the enterprise networking industry by storm, and network automation has been at the eye of the storm for a few years. Enterprise businesses and organizations that are spending millions of dollars on traditional IT lifecycle management based on traditional IT frameworks have been searching for a new IT framework that will provide more stable and predictable network operations without interruption to their IP service and end-user applications. Quite a few network engineers have already started exploring network automation, and the remaining engineers are still trying to make a start. Still, many are encountering various challenges. Looking back on my own networking automation journey, I know it can be a slow, painful, and daunting experience.

Whether you have never touched a programming language in your career or are new to software-defined networks, starting on the network automation journey feels like you are trying to climb up a big mountain.

Once you reach the first peak, you may realize that even bigger mountains await. You may even want to give up.

You probably have been living a comfortable life as a network engineer for the past few decades. Now your bosses expect you to upskill your nonexistent programming skills so you can replace manual tasks with lines of code so that you can add more value to your company. If you say no, your bosses have a Plan B: somebody else will write the code for your team to automate your work, and you will start losing credibility (and maybe even your job). You need to step outside of your comfort zone to start on our network programmability journey. Network programmability still is uncharted ground for most traditional network engineers.

In this chapter, we will identify three main IT domain groups in a typical IT environment today. Then, we will define the common IT skill sets that each IT domain group possesses to determine each group's relational strengths and weaknesses. This book was written by a network engineer for network engineers and reviewed by a Cisco Network Academy instructor. The strengths and weakness discussions will be from an enterprise network industry perspective. The point is to learn how the Network group can use realistic learning strategies to get closer to the other two groups (DevOps and systems groups) and grow into cross-functional engineers who have strong networking skills as well as the skill sets to manage Linux operating systems and write code to develop business applications. This chapter also introduces a cross-functional "hybrid engineer" concept, which will soon be in enormous demand in the IT job market. The career knowledge growth for such hybrid engineers will take the shape of a *T*, as they have a firm foothold in their main IT domain and extend out to other domain skill sets as the top of the letter *T*.

In almost every chapter of this book, except this one, you will install, configure, practice, code, and learn from your PC or laptop, so your system's minimum requirements are the success of your Python network automation study with this book. Hence, we will review the minimum system requirements for your system and introduce you to the software used to gauge the size of crucial tasks outlined in this book. We will also briefly touch on an integrated development environment (IDE) for Python network automation development and provide all the download links for the software, source code, and files used in this book. At the end of this chapter, you will better understand a current network engineer's strengths and weaknesses. Hence, you will know the gaps (target study areas) you have to focus on to start writing Python code to develop your own network automation applications.

Exploring Your Skills and Prerequisites

The section discusses the three main IT domain groups' skill sets at work today. You will learn each group's strengths and weaknesses. From the network group's perspective, we will discuss and dissect the weaknesses in your skill set and develop workable study strategies to work on the weaknesses and turn them into your strengths. Studying Python syntax and concepts will help you reach 25 percent of your network automation

goals. What about the other 75 percent? To write Python network automation code, you have to become strong in many other areas besides networking, and this chapter will help you become better in these areas. Of course, we will also cover Python basics from a networking perspective. Together, we will work on a study plan to address the common network engineer's weaknesses and guide you in the right direction for network automation using Python.

If you are currently working in the IT industry, especially on enterprise-level routing, switching, and security technologies, you should take pride in your job. You probably can relate to the image presented in Figure 1-1. Although the image might make you laugh, you have probably seen an IT engineer who walks and talks like he knows it all. We have all been in that situation where our clients expect us to know everything as the technical expert.

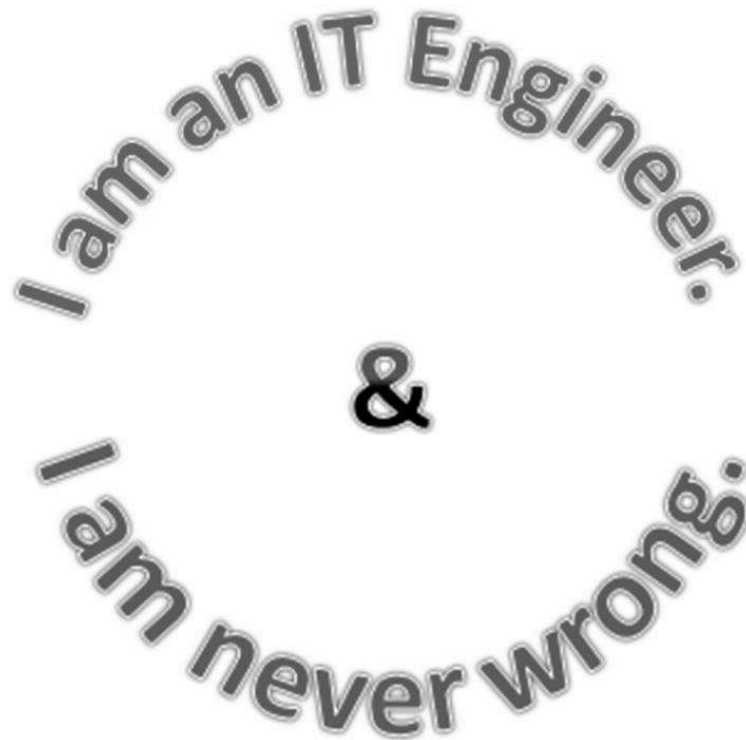


Figure 1-1. IT engineers

Sadly, the image in Figure 1-1 holds some truth to how some IT engineers think and behave at work. Since the first personal computer was invented in 1971, many IT jobs have come and gone as different technologies have emerged and disappeared. Most enterprise IT ecosystems continuously evolve as new technologies are introduced into the IT market. Unfortunately, some IT engineers are too stubborn to move with the times and often get caught out during technology transitional periods, which results in early termination from their IT careers. Currently, the age of artificial intelligence (AI) and IT automation has arrived in the enterprise network IT industry. If you refuse to skill up, your position might not be as secure as you think.

In fact, for many years, the IT industry has been trying to drive the operational costs down by offloading less-skilled jobs to developing countries where the IT operational cost is relatively cheaper than on the home turf. For many years, driving the operational costs down and reducing the

overheads spent on human resources were trends for many organizations. We live in an era where human resources are an overhead cost in a business model, and many clients these days want to pay less but still demand quality IT delivery services. Contrary to many organizations' claims, many local IT jobs are considered overhead rather than valuable human resources for an organization in advanced countries. That is, many IT organizations claim they value their staff, and their priority is the well-being of these human resources in the organization's IT operation. Still, when the crunch time comes, almost every organization succumbs to the financial pressure and cuts IT operational costs to maximize the organization's profit. In the last 20 years, IT outsourcing and offshoring efforts have had great success to some extent. Still, each IT department is driven harder than before to drive the operational cost down, which has been the catalyst in speeding up the adoption of IT automations such as software-defined networks and infrastructure as code.

General Competencies of Three Main IT Domain Group

Bangalore, India, is commonly known as the Silicon Valley of India, and there is a common saying there: "There are two types of people in IT profession; one, IT professionals, and two, professionals who manage these people." In this book, to help your understanding, we are dividing IT domain groups into three different groups based on each group's competencies and characteristics. Then we'll compare their general technical competencies to each other to forecast what the near-future IT industry may look like for one particular group of engineers, that is, the network engineering (connectivity) group. Who will be at the forefront of the networking field in the next five years and beyond? Let's review the gaps and requirements to look ahead and develop a plan to study Python, network automation, and any other requirements to build your confidence to skill up in IT.

At the enterprise level, many organizations have three main domain groups that look after IT infrastructure.

- *Network*: The network group, sometimes known as connectivity group, looks after all infrastructures and end-user connectivity.
- *Systems*: The systems group looks after servers and applications to provide service to other infrastructure or end users as service; these will include corporate emails, shared storage, and various business applications critical to the success of the company.
- *DevOps* : Last is the group of software engineers or programmers who test, develop, and implement various business applications to cater for a company's business requirement. Recently these group of engineers have become known as DevOps engineers.

We will refer to these three main domain groups as *network group*, *systems group*, and *DevOps groups* for the simplicity of this book. The network group works on IP connectivity and services, including technology subdomains such as RS, security, data center, cloud, and unified collaboration engineers. A network connection is the foundation of any enterprise business connectivity, and we consider both applications and services as the tenants of connected networks. Hence, the entire business

comes to a halt if there is a major network outage on the corporate network.

The systems group is responsible for critical business applications and operating systems running on both Windows and Linux OS.

The DevOps group comprises software developers and programmers who specialize in developing software and applications using various programming languages and software development tools. By comparing and analyzing each group's current capabilities, we will better understand each group's strengths and weaknesses. Hence, we identify several weaknesses in the network group so that you can work on these weaknesses. Although the networking vendor technology used in this book is from Cisco, note that the network automation concepts shown in this book will apply to any vendor networking and security technologies including Cisco, Arista, Juniper, Check Point, Palo Alto, and HP.

First, let's compare what technical capability differences exist among these three domain groups. To analyze each group's competencies, we will use a spiderweb graph to plot and illustrate different technical capabilities. After this simple comparison, we will recommend on ways to embark upon network automation using Python as the preferred programming language.

To easily compare and help your understanding of the three IT domain groups' capabilities, let's first plot each group's competencies on a graph, as shown in Figure 1-2. The graph uses ten scores for all competencies plotted on the graph. For example, a competent level in an area would be eight out of ten, and a less capable person would get a score of two out of ten.

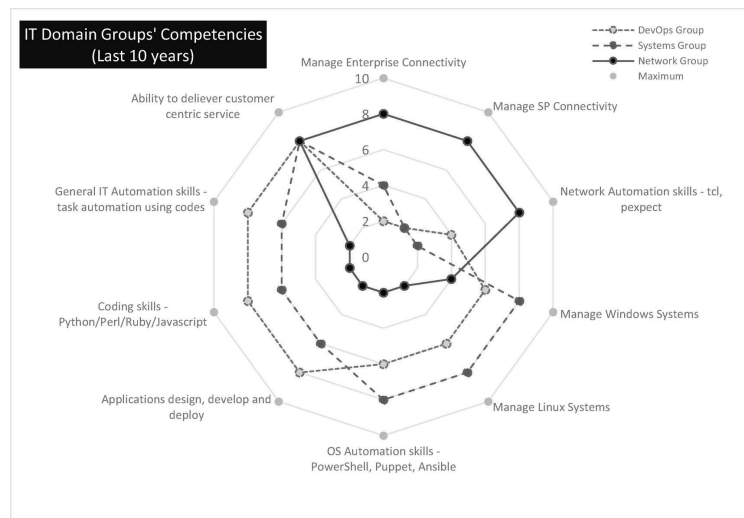


Figure 1-2. IT domain groups' general competencies comparison

The graph represents the common competencies required by each engineering group in the past ten years. Plotting the common competencies on such a graph already gives us an eagle-eye view of where each group's strengths and weaknesses exist. Traditionally, network engineers did not have a strict requirement to learn software engineering or write code as the work pace was a lot slower than today's networks need. As discussed, most of the work has been done by conventional network engineers sit-

ting in front of their computer, through command lines. The system group has been exposed more to system automation tools and languages because of the work as they have to perform most tasks in batches (bulk). From a network group's perspective, writing code to automate the enterprise network was a new concept until five years ago. Hence, network engineers had no urgency to expand their skill sets toward the systems or DevOps domain. For someone to automate an enterprise network, one must possess a combination of all three groups. None of the three groups has all the competencies to achieve enterprise network automation from the graph. And it would be safe to guess that you would have minimal exposure to managing enterprise systems or be involved in enterprise tools (application) development projects at work if you are coming from a conventional network engineering background. In recent times, everyone is expected to deliver more bang for the customer's buck and with a good customer (end-user) experience.

This same mentality has caught up to the work of network group. Still, you cannot throw in more bodies to deliver the service levels as in the offshoring method or cut corners to make up the numbers, as many sales operations promise more but deliver the bare minimum. Also, as the graph suggests, most companies now realize that there is no single group with all the competencies to deliver an enterprise network automation solution for the company. The company does not want to pay for the overhead of running all three IT domain groups; hence, the current focus is to upskill the existing employees to achieve IT automation for the company. IT business insiders have been saying, "The walls of different IT domains are crumbling." Now, network engineers must know how to administer a Linux operating system, have the skills to manage a programming language, and must know how to develop a specific application to automate the enterprise network they manage. On the flip side, any engineers from the systems or DevOps groups can learn networking technologies and automate the network group's work.

Of course, the graph contains the generalized capabilities of each IT domain group, and the competencies of each engineer in each group will vary from one person to the next. Still, this graph provides us with accurate pointers on where and how a network group approach should approach Python network automation studies.

Comparative Analysis of IT Engineers' Responsibilities

In Figure **1-2**, you saw the competencies of the three IT domain groups. In this section, we will look at the typical responsibilities of engineers from each group to pinpoint the areas a network engineer has to improve to embark on enterprise network automation using Python. Spend about five minutes studying the responsibilities of each group, and you can easily see where a conventional network engineer must focus to gain the required skills to write a scripted application to automate their work. As the shaded area shows, the first requirement is mastering operating systems such as Linux and Windows; this is on top of strong networking skills.

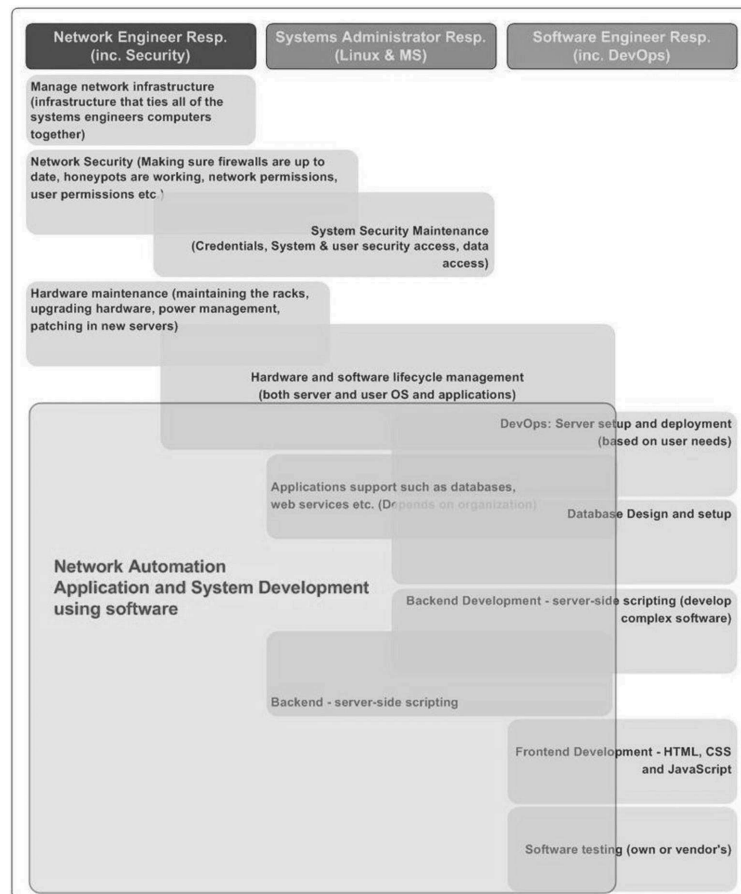


Figure 1-3. IT engineers' general responsibilities

For enterprise automation, the default operating system used is Linux. After getting familiar with basic to intermediate Linux administration, you can start with basic Python syntax and slowly discover how DevOps work. At the same time, you need exposure to Linux grep or, even better, general regular expressions, as data processing is the key to any programming language. From this point on, you can gradually expand to other areas of interest.

As mentioned, you are expected to have a working knowledge of Cisco CCNA to get the most out of this book, so if you are studying for or have already completed CCNA, you will get a lot more from this book. Although the content of this book does not teach general networking concepts from Cisco Systems, much of the book is very hands-on, and the labs in later chapters are based on Cisco's IOS and IOS XE configurations. Even if you are not a certified Cisco Certified network engineer, I base this book's content on the discussions in Figures 1-2 and 1-3, so the topics are broad with many helpful parts to upskill your general IT skills.

Based on Figure 1-3, let's draw on the skill sets a network engineer needs to get ready for network automation using Python. The following are derived from the responsibilities of systems and DevOps engineers, but not the core responsibilities of a typical networking job:

- System security maintenance (credentials, systems, and user security access control, data access control)
- Hardware and software lifecycle management (both server and user OS and OS-specific applications)
- DevOps, server setup, and deployment

- Applications support such as databases and web services
- Database design and setup
- Back-end development server-side scripting (developing complex software)
- Back-end server-side scripting
- Front-end development with HTML, CSS, and JavaScript
- Software testing (your own or vendors’)

The previous responsibilities are quite general and broad. Many subdomain technologies are not even mentioned in the previous list. But from a network engineer’s perspective, while having a firm footing in network foundational responsibilities, you should realize that you have to shift your interest and skill sets toward systems and DevOps groups’ responsibilities. To make your life easier for you, see Table 1-1. As mentioned earlier, this book is not a network core concept book, a Python syntax and concept book, or a Linux core concept book. This book is a combination of three books, but the topics are chosen selectively to get you ready for Python network automation. The skills you will learn in this book will serve as the primer for your knowledge.

As you go through each chapter, you will gain new IT skills outside of your comfort zones. You can see that the technical approach shown in Table 1-1 is practical, including installing virtual machines, installing network services, learning Linux and Python basics, installing Python, and controlling networking devices by writing Python code. Unless you are a presales engineer or an architect for your IT domain, you need to be a true professional who can do the real hands-on work. This table shows one way a network engineer can approach their first journey into Python network automation.

Table 1-1. Skill Recommendations for Network Engineers Embarking on Python Network Automation

Required Skills	Recommended Topics to Study
Python basics	Install Python on your computer Learn basic Python syntax Relate and apply basic Python syntax to a Python network context
VMware and virtualization basics	Install the latest VMware Workstation Pro version Learn virtualization basics Learn to build virtual machines
Operating system management, patching, upgrading, troubleshooting, and configuration skills	Install two different flavors of Linux, Ubuntu 20.04 LTS and CentOS 8.1 Use a VMware template to install a prestaged server; import GNS3 VM Server Read the Python installation guide and troubleshoot any installation problems
Learn Linux basics	Understand the Linux directory

Required Skills	Recommended Topics to Study Understand Linux file management and practices Learn how to use the vi and nano text editors without a GUI desktop app GUI Desktop Linux remote connection concepts, SSH, Telnet, basic APIs Install the FTP, SFTP, TFTP, and NTP servers
Regular expressions	Learn regular expression basics Learn how regular expressions are used in Python Learn to relate and apply regular expressions to networking ideas and concepts
Network emulator (GNS3) skills	Learn to install and integrate GNS3 Learn to configure basic to complex network topology Learn to integrate GNS3 with VMware Workstation
Cisco IOS and IOS XE command lines	Learn to integrate Cisco IOS, IOS-XE, and CML-Personal L2 and L3 images into GNS3 Learn basic router and switch commands for the network automation lab setup Write Python code to control emulated Cisco network devices via SSH, Telnet, FTP, and SNMP Practice Python scripting in a virtual lab Install and set up virtual routers and switches with Python scripts
Application development and establishment	Install Python 3 Install Python Telnet and SSH modules Understand application development for network automation with Python code Understand basic DevOps development concepts
Specific engineer's task automation	Understand the task flow to automate specific tasks

The recommended topics to study from Table **1-1** form the basis for this book. At various points of this book, you will be introduced to different technologies and exercise-based tasks, which require you to follow the book's content in front of your PC (or laptop). To complete this book in full, you must complete all exercises in each chapter to get enough practice and build various skills required for network automation using Python. I want you to learn the basics in one place and learn them well.

Python and Network Automation Studies

Why do you want to study Python for network automation? This question might be tricky to answer. The answer may vary from one person to the next person. Still, perhaps the common goal in studying Python for network automation is to convert physically repetitive tasks into code lines so that you can focus on more valuable and important tasks at work. The process of network automation using Python could be very different from the process of network automation using Ansible or other device management tools. You have the freedom to structure your thought and task processes while trying to automate actual physical tasks and thinking processes targeted for automation. So, what does it mean to streamline the thought process? Perhaps you are talking about performing your work more effectively, systematically, and quickly using Python code.

Are you hinting that what you and your team have been doing all these years has been inefficient, slow, and unstructured? Maybe that is why you are here and reading this book. If so, first review the currently inefficient work your team is performing and break each logical thinking process and physical task performed into steps. It can be extremely handy to determine what you are trying to automate with lines of Python code. This will give you a great foundation to start your first network automation project and approach your current automation challenges.

Rather than embarking on your Python studies, look around and identify the manually intensive tasks at work that you want to automate and then document current processes. Now, you can start learning Python syntax. This simple process will keep your motivation high and also keep you focused on your first Python studies because you are not studying Python for the sake of learning one of the most popular programming languages at the moment. You are learning it for a purpose.

Why Do People Want to Learn Python?

Python is used by leading IT service vendors and enterprises such as Google, YouTube, Instagram, Netflix, Dropbox, and NASA. There are many areas where Python programming can be developed in a short time and applied to solve actual problems. The barrier to learning it is one of the lowest out of all the major programming languages in use today.

According to [RedMonk.com](https://redmonk.com), in June 2020, Python is the second most popular programming languages globally, overtaking Java and gaining momentum. There are pros and cons of each programming language, and Python is not immune to various cons, but why is it becoming so popular?

In October 2019, Python even overtook JavaScript as the most asked for languages on Stack Overflow.

So, how do you begin? You must master the basic Python syntax and concepts well. There are many books on Python basics and concepts, some of which are outstanding, but I still could not make the connection between Python concepts and network automation concepts when I was learning. Strangely, there were also intermediate to advanced Python networking automation books, but they did not explain where to start and what drills to do to reach the intermediate to advanced level. Everyone tells you that Python is such an easy language to learn and the barrier to entry is low, but they do not tell you that if you want to become great at any programming language, you need to put your head down and be persistent and passionate about the language you have chosen. Be ready to give up your social nights and weekends to learn some silly Python library features and watch hundreds of hours of so-called Python gurus' YouTube videos and online training videos.

That is correct; a low barrier to entry does not mean that Python is the most straightforward language to study. It means that anyone can start learning Python as a programming language, but how many people are persistent and passionate enough to make writing Python code their life? The most challenging part of learning Python often is to be persistently pushing yourself to learn different ways of doing things using Python and keeping your passion alive as long as you can. To do this, you have to find practical use cases and small projects that are personal and that matter to you and your work. Remember that automation is a slow and steady process as everything you want to automate must be written into lines of code. Somebody or something (in the case of AI) has to write the code. There is no AI good enough to imitate an experienced and logical network engineer. Hence, the network engineer has to automate tasks until the artificial intelligence (AI) can imitate and automate the network engineer's actions.

In summary, Figure **1-4** is from a 2019 Stack Overflow survey. Python is the fastest-growing programming language and the most wanted language for the third year in a row, meaning that developers who have not yet used it say they want to learn Python to develop applications.

Most Loved, Dreaded, and Wanted Languages

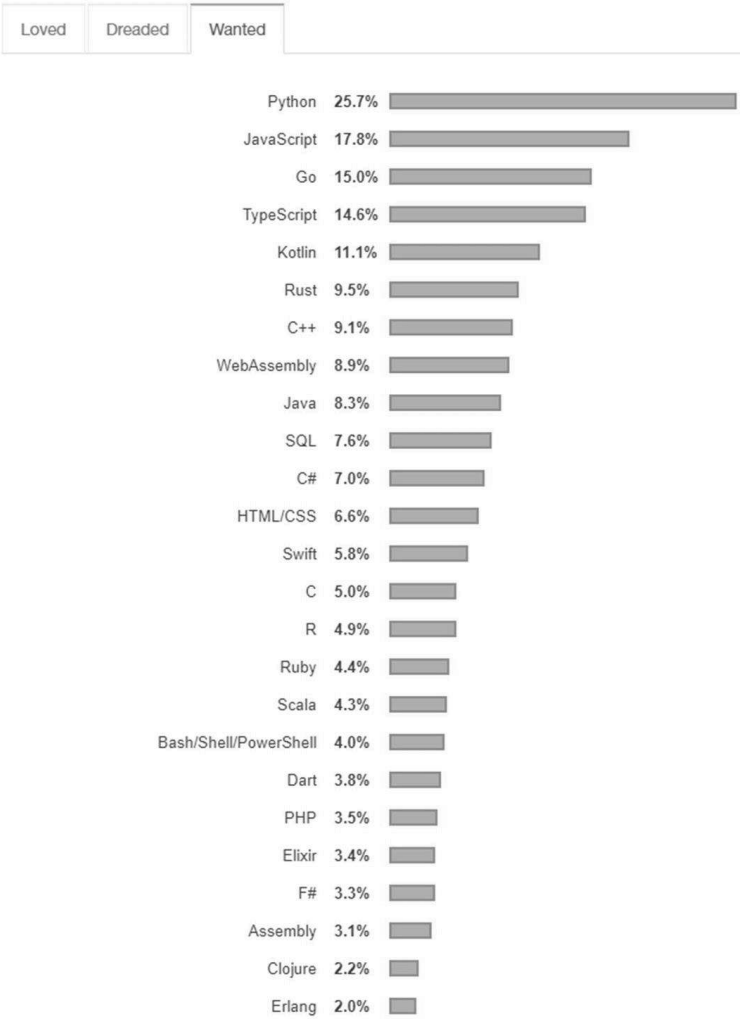


Figure 1-4. Most wanted programming language for developers in 2019 Source: <https://insights.stackoverflow.com/survey/2019#most-loved-dreaded-and-wanted>

Figure 1-5 shows the growth of major programming languages in 2017.

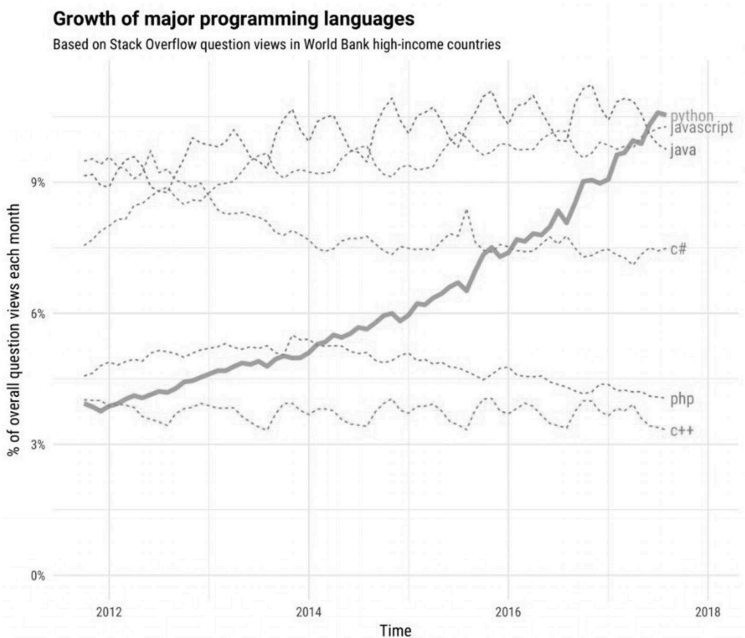


Figure 1-5. Growth of major programming languages based on Stack Overflow questions, 2017 Source: <https://stackoverflow.blog/2017/09/06/incredible-growth-python/>

What Do I Need to Study Network Automation Using Python?

Most network engineers who have been studying network automation will agree that having a flexible integrated development environment (IDE) will help engineers develop applications. There are many ways you can study basic Python syntax, but not everyone has the comfort of the company's provided network automation development lab environment, although Cisco dCloud or similar online sandboxes offer timed lab environments. Still, it is not exactly available to you on-demand around the clock. So, I recommend you use your own equipment to control your lab. Such a lab environment can be configured in several forms. First, it can be configured with 100 percent hardware using routers, switches, and servers. Second, it can be part hardware and part virtualized, a hybrid development environment, or third, it can be 100 percent of equipment virtualized and live inside a server or a PC. The first method will require a large initial investment to purchase second-hand equipment and incur ongoing electricity bills to keep the physical devices operational. A hybrid development environment will become cumbersome to have some devices run on physical devices, and some run on a virtualized environment. The last environment would be an ideal lab environment for anyone studying Python for network automation. This has been made possible by integrating several systems and network operating systems running on a virtualization solution. It is almost impossible to practice and experience everything that occurs in the real network environment, but building the integrated development lab will be the closest thing you can get to. If you are learning everything through books or passively learning Python network automation by binge-watching YouTube videos, that is not good enough. Build your IDE and type your code to learn and master Python.

To study Python network automation through this book, all you need is a reasonably powerful desktop or laptop computer that has enough central processing unit (CPU) power and enough random access memory (RAM). Next, let's discuss the minimum hardware specifications to follow this book.

Hardware: Minimum Specifications for Laptop or PC

To install, configure, and practice all the exercises and Python network automation labs in this book, your computer must meet or exceed the minimum specifications listed in Table 1-2. Since most network engineers at work use Windows 10, we will also use this OS as the base operating system for building our IDE lab environment. Unfortunately, if you are using macOS or Linux, then you are on your own with finding the right software and compatibility. If you have a powerful laptop or PC with Windows 10 freshly pre-installed, you are in a perfect place to read this book, but if your system is still running Windows 8.1 or 7, it is highly recommended that you upgrade to the latest Windows 10. If you have a reasonably powerful desktop PC with optimized CPU and system cooling, it will run much better than a laptop. However, if your laptop is the latest

and greatest, I would recommend using your laptop for mobility and a lower electricity bill.

Since a large part of this book focuses on installing software, creating a practical lab, and running network automation labs on a single Windows 10 system, make sure your system meets the minimum requirements specified in Table 1-2. Otherwise, you will encounter system performance issues such as slow system response time in your lab.

Table 1-2. Laptop or PC: Minimum Specifications

PC Components	Minimum Specification
CPU	Intel: CPU i7 Gen3 (64-bit) or above AMD: Ryzen 3600 or above
RAM	16GB or more for DDR4 (or 24GB for DDR3)
SSD	240GB or more (with 15 percent or free space for system paging)
OS	Microsoft Windows 10 (64-bit) or newer

Figure 1-6 shows the desktop system I have used to write this book.

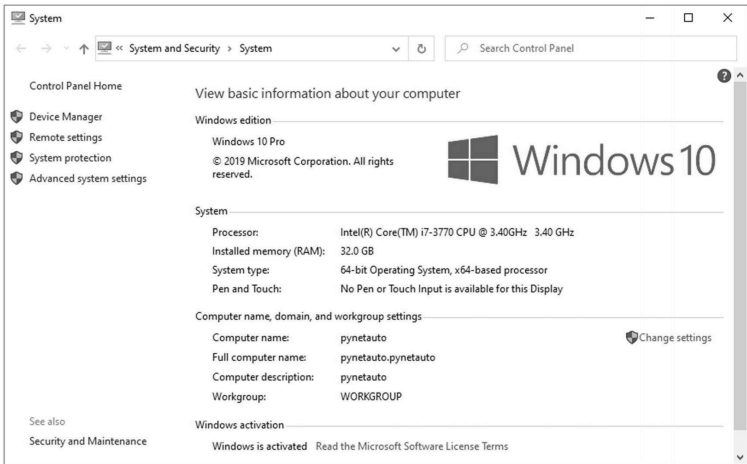


Figure 1-6. My system specs

If your CPU is on par or better than Intel i7-3770, as shown in Figure 1-6, and has more than 16GB DDR4 (or 24GB DDR3) RAM, all labs should run smoothly. Also, SSD is recommended over HDD because the mechanical hardware is one component that can become the system bottleneck. This book aims to enable you to create a hands-on lab where you can study Python, virtualization, Linux, networking, and network automation, all from a single PC/laptop. As we will explain later, the labs in this book are useful for Python network automation study and those studying for various certifications such as Cisco CCNA/CCNP/CCIE, Checkpoint, Palo Alto, and Juniper Junos, as GNS3 supports various vendor OSs.

Software Requirements

You will learn how to install and integrate various technologies while building a highly flexible and integrated lab on a single PC/laptop. Table [1-3](#) contains all the software and provides the download links. You can download all software before beginning Chapter [2](#) or follow along with the book and download different software as instructed at the beginning of each chapter.

Please note that not all software used here is freeware or open source software, and you may have to use demo software or paid software. For example, VMware's Workstation 15/16 Pro will run in demo mode for the initial 30 days, and then you will have to purchase a valid key. Another example is Cisco Modeling Labs-Personal Edition (CML-PE), which has a subscription of \$199 yearly. For our lab, you are only going to require the three files outlined in Table [1-3](#). All other software used here is either freeware or open source software.

Table 1-3. Software and Download Links

Required Software	Usage	Download links
VMware Workstation Pro 15 or 16 (demo/paid)	Desktop Virtualization	https://my.vmware.com/web/vmware/downloads/ or https://my.vmware.com/web/vmware/downloads/
GNS3-2-2-14-all-in-one or later (for Windows)	Network device emulator	https://github.com/GNS3/gns3-gui/releases or https://www.gns3.com/software/download
GNS3.VM.Vmware.Workstation.2.2.14.zip	VMware Workstation GNS3 VM ova image	https://github.com/GNS3/gns3-gui/releases
IOSv-L3-15.6(2)T.qcow2 or newer (paid)	Cisco CML L3 image	https://learningnetworkstore.cisco.com/cisco
IOSv_startup_config.img (paid)	Cisco CML L3 booting image	https://learningnetworkstore.cisco.com/cisco
IOSvL215.2.4055.qcow2 or newer (paid)	Cisco CML L2 image	https://learningnetworkstore.cisco.com/cisco
python-3.8.5.amd64.exe or newer (for Windows)	Python 3.8.5 or higher	https://www.python.org/ftp/python/
Notepad++ v.7.8.5 (64-bit)	Text editor for Windows	https://notepad-plus-plus.org/downloads/
Putty.exe (64-bit) (for Windows)	SSH/Telnet client	https://www.putty.org/ or https://the.earth.li/~sgtatham/putty/0.74/w

Required Software	Usage	Download links
Ubuntu-20.04-live-server-amd64.iso (952.1MB)	Ubuntu Server image - Bootable	https://linuxclub.cs.byu.edu/downloads/isos or https://www.ubuntu.com/download/server
CentOS-7-x86_64-DVD-1804.iso (4.16GB)	CentOS Server image, bootable	https://www.centos.org/download/ or http://centos.mirror.omnifrance.com/
c3725-adventerprisek9-mz.124-15.T14.bin or c3745-adventerprisek9-mz.124-15.T14d.bin	Cisco IOS im- age for GNS3 integration	From old used Cisco devices or Through Google search
Unpack.exe	Cisco image unpacker 0.1 binary for Windows	http://downloads.sourceforge.net/gns-3/Unpack.exe
WinMD5Free	Windows MD5 Checker	https://www.winmd5.com/
WinSCP	Secure File Transfer for Windows- Linux	https://winscp.net/eng/download.php
FileZilla client (64-bit, for Windows)	Secure File Transfer for Windows- Linux	https://filezilla-project.org/

Building a Network Automation Development Environment Using GNS3

A network automation development environment is an IDE lab for network application development. For our convenience in this book, we will call this environment the *lab* or the *network lab* instead of an *integrated development environment* or *network automation development environment*. As discussed earlier, there are different ways to create a learning lab for your Python network automation. About 15 years ago, while networking students studied Cisco routing and switching, many students used a mix of hardware-based labs, a networking OS simulator from Cisco called Packet Tracer, and an open source emulator called Dynamips. This was followed by the introduction of GNS3, which is Dynamips with a GUI interface for ease of use. Dynamips was a good networking device emulator used for study, but it lacked the GUI and made it a little challenging for beginners. GNS3 was a program that provided a graphical user interface (GUI) to manage emulated lab devices with ease.

In the early days, GNS3 had a lot of bugs and did not always work well. With the development of more advanced CPU architecture, increased RAM capacity, and the introduction of the solid-state drive (SSD), emulating vendor network devices from an application on your laptop or PC has become more comfortable and more accessible. The price reduction of PC hardware components and more open source network emulators made studying networking extremely accessible. The latest versions of GNS3 are free from many pre-existing bugs and can run a very stable lab environment. GNS3 has evolved to support older Cisco IOS and integrate Cisco IOU and CML's (VIRL) L2 and L3 IOS. You can run both L2 switches, and L3 routers can be emulated and run in a virtualized lab environment. Although the best way to study any technology is to use the proper equipment or the same virtual infrastructure, this option might not be available to everyone. If both the money and time are available, it is best to study with proper equipment. It is always better to study with an emulator than with a simulator and is even better if you can afford to run the real lab using the proper equipment. Table 1-4 provides a tabulated list of applications used for network labs to help you with some history of network simulators and emulators for networking studies.

Table 1-4. Network Emulation/Simulation Program for Networking Studies

Application	Simulator /Emulator	Free/Open Source/Proprietary	Pros and Cons
Cisco Packet Tracer	Simulator	Proprietary	Pros: Free to Network Academy students Very easy to use (suits beginners) Useful for learning the basic concept Readily available on torrent sites or free file share sites (download and use at your own risk) OK study tool for Cisco Certification Cons: Nothing close to real Cisco IOS Nothing close to real Cisco hardware
Dynamips	Emulator	Open source (still requires extracted Cisco IOS image(s))	Pros: Free Reasonably easy to use (suits in-

Application	Simulator /Emulator	Free/Open Source/Proprietary	Pros and Cons
			intermediate users) Useful for learning the basic concept Support for older router platforms Very versatile Easy integration with virtual machines Easy integration with other hardware Fewer bugs than GNS3 Support for both Windows OS and Linux OS Excellent study/work tool for Cisco certification Cons: You'll have to find your own IOS copies Only support older IOS routers (except for 7200 routers) Limited switching capabilities Support through open source forums only
GNS3	Emulator	Open source (still requires extracted Cisco IOS image(s))	Pros: Free Very easy to use (suits beginners) Useful for learning the basic concept Support for older router

Application	Simulator /Emulator	Free/Open Source/Proprietary	Pros and Cons
			platforms Very versatile Easy integration with Virtual Machines Easy integration with other hardware Support for both Windows OS and Linux OS Perfect study/work tool for Cisco certification
			Cons: Locate your own IOS copies Only support older IOS routers (except for 7200 routers) Limited switching capabilities Lots of bugs Lots of performance issues Support through open source forums only
Cisco IOU	Emulator	Proprietary	Pros: Reasonably easy to use (suits intermediate users) Support for later or latest IOS Excellent L2/L3 switch emulation Can run on most WIN/Linux hosts Readily available on torrent

Application	Simulator /Emulator	Free/Open Source/Proprietary	Pros and Cons
			sites or free file share sites Excellent study/work tool for Cisco certification Cons: Proprietary Requires a relatively powerful host machine No support from Cisco. Find your copy and use at your own risk Pros: Fairly easy to use Support for later or latest IOS Excellent L2/L3 switch emulation Can run on most WIN/Linux host Excellent study/work tool for Cisco certification Support from Cisco if paid for subscription Readily available on torrent sites or free file share sites Cons: Proprietary Requires a relatively powerful host machine
Cisco CML (VIRL)	Emulator	Proprietary	Pros: Fairly easy to use Support for later or latest IOS Excellent L2/L3 switch emulation Can run on most WIN/Linux host Excellent study/work tool for Cisco certification Support from Cisco if paid for subscription Readily available on torrent sites or free file share sites Cons: Proprietary Requires a relatively powerful host machine
UNL Lite (EVE-NG)	Emulator	Proprietary	Pros: Support for later or latest

Application	Simulator /Emulator	Free/Open Source/Proprietary	Pros and Cons
			IOS Excellent L2/L3 switch emulation Can run on most WIN/Linux host Excellent study/work tool for Cisco Certification Cons: Requires a relatively powerful host machine Pros: Free Tool for building and managing virtual machine environments in a single workflow Easy-to-use workflow Focus on automation Lowers development environment setup time Increases production parity Cons: Requires a relatively powerful host machine
Vagrant		Open source	

Other emulators introduced in the last decade include Unified Networking Lab's UNL Lite (renamed EVE-NG), Cisco's IOU (IOS On Unix), and Cisco's CML (Cisco Modeling Lab, old VIRL) . Cisco Packet Trace can be used while studying Cisco certifications using Cisco Network Academy account. The software was used for experimentation only within the Cisco Technical Assistance Center (TAC) with Cisco IOU. It cannot be used by companies or individuals other than Cisco TAC. There have been many books and free guides on the Web on how to use the previous software, but in this book, GNS3 VM running on VMware Workstation Pro, integrated with Cisco IOS and CML images, will provide us with the emulated network infrastructure for Python network automation labs.

For this book, the ideal lab is the one that is optimized for less power consumption, which can be operated conveniently at any time and anywhere. So, this book has carefully laid out the topics in sequence, from easy to intermediate. Also, as long as you use the same or newer versions of software used in this book, most of the lab setup should work without software compatibility issues, and all labs should be reproducible.

Suppose you want to use this book's content for group training and only want to teach students in Python labs. In that case, you could install Windows 10 as a virtual machine on ESXi 6.5 (Figure 1-7), install all software, complete the integration, and then make clones of the original virtual machine and provide RDP accesses to designated virtual machines.

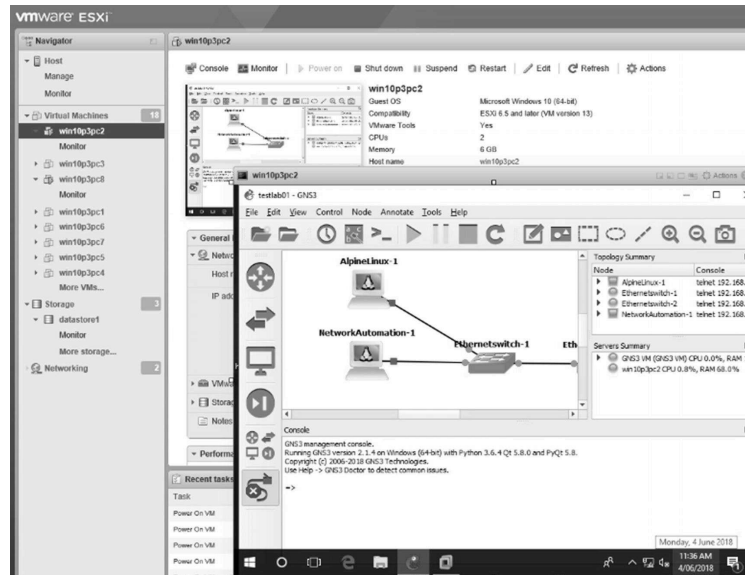


Figure 1-7. Example of running a network automation lab on ESXi 6.5 for group training

Downloading Supplementary Guides and Source Code

Download all pre-installation guides and source code before reading Chapter 2.

Supplementary pre-installation guides download:

- Author's URL:
https://github.com/pynetauto/apress_pynetauto

Source code download:

- Author's URL:
https://github.com/pynetauto/apress_pynetauto

Summary

We discussed IT network automation in general and did a quick comparative analysis of three main IT domain groups. We identified areas that the network engineer group must focus on to start writing Python code for network automation application development. Finally, we discussed the

minimum hardware and software requirements to embark on your network automation journey. You probably noticed no networking hardware requirement is mentioned as part of the hardware requirement; everything will be software-based, even the routers and switches.

Storytime 1: Mega Speed Shoe Factory



In 2016, there was a great hype around robotics and automation replacing human labor in the shoe manufacturing industry. A famous shoe and sports apparel manufacturer, Adidas of Germany, announced a new robotic shoe factory; during the announcement, there was a prediction that up to 90 percent of 9 million workers in Southeast Asia could face unemployment because of this type of factory. However, in 2020, Adidas announced it would halt production at its two robotic “speed factories” in Germany, Ansbach, and the U.S. factories. The robotic shoe manufacturing line could only produce a limited number of models that mainly consisted of running shoes with a knit upper sole; the robots could not produce leather shoes with rubber soles. The old pieces of machinery have been moved and are currently used by Adidas suppliers in Vietnam and China.

Some people say machines will take over the world and take all our jobs sooner than later, but this example amplifies the need for more careful planning and the introduction of workplace automation. Each industry is learning the power of automation through trial and error.

This failure also takes us back to the early 1990s when automated teller machines (ATMs) were introduced. In that case, some studies in the United States have found that the economic forecasts made by many prominent economists and scholars were as accurate as your local taro fortune teller. So, if we are well prepared and embrace the changes, we should not fear automation or AI in the IT industry. While supporting your company’s IT ecosystem and having pride in what you do at work is extremely important to all IT engineers, we have to be truthful about what we can do and what we know. An IT engineer does not know every aspect of all the IT technologies they support. Hence, they should not attempt to cover up their inability to perform specific tasks. We should always be ready to discover new things and gain additional skill sets to advance our careers.